University of Sunderland

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MODULE TITLE: Full Stack Development

MODULE LEADER: David Grey

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ASSIGNMENT: 1

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Live demo Link: https://rihanashrestha567-star.github.io/Demos/html_demo.html

What is HTML?

HTML is the language used to create web pages. It gives the structure and content of a website using a system of tags and attributes. HTML defines things like headings, paragraphs, lists, links, images, forms and many other components that browsers turn into pages.

Modern HTML5 introduced elements like header, nav, main, section, article, footer that give meaning to different parts of a webpage. These elements improve accessibility for screen readers help search engines understand content better and make code more readable and maintainable for developers.

HTML works with CSS and JavaScript to create complete web experiences. Without HTML there would be no structure for the technologies to build upon.

Why HTML Matters in Modern Web Development

Good HTML is the foundation of every website. Structured HTML leads to accessibility problems, SEO issues and maintenance nightmares. HTML5 has become increasingly important as search engines and assistive technologies rely on markup to understand and present content correctly.

HTML5 also introduced features like native video and audio support form validation, canvas for graphics and local storage. Reducing the need for external plugins and libraries.

How NepalRide Demonstrates HTML

NepalRide uses HTML5 throughout the entire application. Every section of the page is wrapped in elements rather than generic divs. The navigation uses a nav element the main content areas use main and section tags and each car card is structured as a self-contained component.

The Portfolio section includes a dedicated 'HTML Demo' pane that shows a clean semantic car listing example using header, section, ul and footer elements. Demonstrating practices without any CSS applied.

All forms use label-input associations, required attributes and appropriate input types. The mobile menu button includes an aria-label for accessibility.

Code Example 1. Core Implementation

```html
<article>
<header>
<h2>Mahindra Scorpio-N (SUV Package)</h2>
<p><strong>Category:</strong> Sports Utility Vehicle</p>
</header>
<section>
<h3>Technical Specifications</h3>
<ul>
<li><strong>Seating:</strong> 7 Passengers</li>
<li><strong>Engine:</strong> Dual-Stage Diesel</li>
</ul>
</section>
<footer>
<p>Developed for NepalRide, Kathmandu</p>
</footer>
</article>
```

Code Example 2. Interactive Feature

```html
<!-- Semantic navigation structure -->
<header>
<nav id="navBar">
<button onclick="switchTab('home')">Home</button>
```

```
<button onclick="switchTab('login')">Login</button>

<button onclick="switchTab('contact')">Contact</button> </nav>

</header>

< class="page-section active" id="home-section">

<!-- Main content here -->

</main>
```

The NepalRide website demonstrates HTML practices through its clean semantic structure visible across all sections. The Portfolios HTML Demo pane specifically isolates markup so users can see the raw structure without styling distractions.

Every interactive element uses HTML elements rather than generic clickable divs ensuring keyboard accessibility and proper semantic meaning.

Figure 1: Portfolio HTML Demo Pane

Shows a textbook example of markup for a car listing card using article, header, section, ul and footer elements. No CSS is applied, demonstrating clean HTML structure.

Figure 2: Contact Form Structure

The booking form uses form elements with labels, required attributes and appropriate input types. This ensures both accessibility and native browser validation.

Key Features & Technical Implementation Details

NepalRide implements HTML5 elements consistently: header for the top navigation, nav for the menu main for primary content areas section for logical groupings and article for individual car cards. This creates a document outline that assistive technologies can navigate easily.

The project also uses form elements with associated labels, required attributes and input types that trigger appropriate mobile keyboards.

Implemented Features in NepalRide:

• Complete structure using header, nav, main, section, article, footer

Challenges Faced & Solutions Applied

- Challenge: Balancing visual design needs with semantic requirements. Solution: Used elements as containers and applied CSS classes for styling without compromising meaning.

- Challenge: Making the HTML Demo pane clearly demonstrate structure. Solution: Created a white box with serif typography to make it look like a textbook example of HTML.

- Challenge: Ensuring accessibility. Solution: Added aria-label to the hamburger menu button and used button elements throughout.

Proper form labels associated with inputs for accessibility

Personal Reflection & Learning Outcomes

Learning semantic HTML completely changed how I structure web pages. Before this module I would use divs for everything. Now I think carefully about what each element represents and choose the meaningful tag.

The HTML Demo pane in the Portfolio was particularly educational because it forced me to create readable markup without relying on CSS to make it look good. This helped me appreciate the importance of structure.

I also learned that semantic HTML improves SEO and accessibility without any effort. It's simply the right way to build websites.

Learning Outcomes Achieved:

- Demonstrated understanding of HTML5 semantic elements

NepalRide demonstrates strong HTML fundamentals through its consistent use of semantic elements, accessible forms and clean markup. The dedicated HTML Demo pane, in the Portfolio section provides evidence of understanding. This foundation supports the advanced CSS and JavaScript features built on top of it.